

Mentor

A trading tutor and probabilistic forecasting web application — built to make a disciplined trader, not to promise a crystal ball.

A single-user web app that teaches trading from first principles, reads market data and news from across the internet, produces an honest probabilistic read of the market, and enforces strict risk management on every trade.

Version 1.0

Personal build specification

Scope

Single-user web application

Reference market

EUR/USD (forex major)

Prepared

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Read this first. This is a product and engineering plan for a personal, educational decision-support tool — not licensed financial advice, and not a system that reliably predicts price. Markets are close to efficient: by the time public information reaches a retail trader, faster and better-capitalised participants have already priced it in. The goal of this product is to make its user a disciplined, educated trader who loses less and understands why — not to beat the market. Any tool claiming reliable prediction is selling a fantasy.

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01 Executive summary

Mentor is two products wearing one coat: a teacher that makes its user a better trader, and a probabilistic forecasting engine. The teacher is where almost all the value lives; the forecasting is a feature inside it, built with deliberate humility.

The application ingests historical and live market data plus news and economic events from across the internet, engineers them into transparent signals, and produces a **probabilistic lean with an honest confidence level and a plain-language reason** — never a fabricated price target. Around that read sits a strict risk-management engine that sizes every trade from how much the user is willing to lose, a backtester that proves whether any signal has a real edge after costs, and a journal that measures whether the user is actually improving.

The reference build targets a single instrument, **EUR/USD**, on a personal single-user web app. Because there is only one user, a capable (and relatively expensive) language model can be used freely for the reasoning and mentoring layers. The north star: after six months of use, the user is a measurably more disciplined trader who understands the reason behind every number on the screen.

02 Vision & guiding principles

Five principles govern every design decision. When a feature conflicts with one of these, the principle wins.

PRINCIPLE 01 — THE MENTOR IS THE PRODUCT

Forecasting is a feature inside a teaching tool. Every number is a teaching moment; every signal carries its reasoning.

PRINCIPLE 02 — REASONING, NEVER VERDICTS

The app explains *why* it leans a direction and *what would change its mind*. It never simply says “buy.”

PRINCIPLE 03 — RISK BEFORE PREDICTION

Most traders fail on risk management, not forecasting. The risk engine is the foundation and is built first.

PRINCIPLE 04 — PROOF BEFORE TRUST

No signal reaches the dashboard until it has survived honest, cost-aware, out-of-sample backtesting.

PRINCIPLE 05 — HONESTY ABOUT UNCERTAINTY

Forecasts are probabilities with confidence bands. A confident price target with no uncertainty attached is a bug.

What this is — and is not

What it is

- A personal trading **tutor** that teaches from first principles and explains its reasoning.
- A **data aggregator** that turns price history and internet news into a probabilistic, uncertainty-honest read.
- A **discipline engine** that enforces risk rules and journals results so the user can see real improvement.
- A **laboratory** for validating ideas through rigorous backtesting before any money is risked.

What it is not

- Not a reliable price predictor. No product is.
- Not an automated money machine or a get-rich tool.
- Not a source of licensed financial advice.
- Not an auto-trader — it advises; the human decides and confirms every action.

REALITY CHECK

Higher predictability and higher profitability pull in opposite directions. The smoothest markets are smooth *because* they are efficient and hard to beat; the markets with exploitable patterns are choppy and risky. There is no instrument that is both very predictable and very profitable. The app is designed around that truth, not against it.

04 Who it is for & how it is used

Primary user: a single self-directed trader who wants to learn properly and trade with discipline, comfortable using a personal web app and willing to paper-trade before risking capital.

A day in the life

- **Morning briefing.** Open the dashboard: overnight price action, today's probabilistic lean with reasoning, the news read, and any high-impact economic events scheduled. The mentor highlights what matters and why.
- **Pre-trade discipline.** Before any trade, the risk engine sizes the position from the user's risk rules, and a pre-trade checklist must be completed — forcing a reason, a stop, and a target on every entry.
- **In-trade.** Alerts fire on price levels and approaching news events. An "event freeze" warns before entering near a high-impact release.
- **Post-trade journaling.** Log the outcome, tag any mistakes or emotions, and let the app update the user's running expectancy and R-multiple distribution.
- **Weekend review.** The mentor reviews the week's journal, surfaces behavioural patterns, and serves the next lessons in the curriculum.

05 Market & instrument selection

The reference build trades **EUR/USD**. The reasoning a seasoned trader would give: the deepest free historical data and the tightest spreads (so trading costs do not distort backtests), clean and well-defined mechanics (pips, lots, position sizing) ideal for teaching, and high liquidity with few ugly gaps.

The honest trade-off is that forex majors are highly *efficient*, so genuine edges are small. A market with more exploitable inefficiency — a single large-cap stock or a stock index where long-horizon trend and the equity risk premium are documented effects — is a reasonable later addition. The instrument is therefore a **configuration value**, not hard-coded, so new markets can be added without a rewrite. But one instrument is mastered first; a mentor that covers everything teaches nothing well.

06 Feature set

Features are tagged by release priority: **MVP** first usable version, **v1** the complete product, **v2** advanced enhancements. The build order is deliberate — foundations and validation before prediction.

A • Mentor & education

Feature	Priority	Description
Adaptive curriculum	MVP	A structured lesson tree from market basics to backtesting and psychology, with progress tracking.
Contextual explainers	MVP	Tap any metric (RSI, position size, R:R, confidence) for a plain-language explanation computed from the live values.
"Why did you say that?"	v1	Every signal expands into its full reasoning chain: the factors, their weights, and what would change the call.
Socratic mode	v2	Instead of giving the answer, the mentor asks the user to reason first, then critiques the reasoning.
Spaced-repetition quizzes	v2	Knowledge checks that resurface concepts on a forgetting curve to make learning stick.

B • Market data

Feature	Priority	Description
Historical OHLCV	MVP	Full multi-timeframe history (1m / 5m / 1h / 1d), cleaned and timezone-aligned to UTC.
Live quotes	v1	Streaming or polled live price during market hours.
Data-quality monitoring	v1	Detects gaps, bad ticks, and stale feeds, and flags them rather than silently corrupting models.
Multi-timeframe view	v1	See the same moment across timeframes — essential context the mentor teaches the user to read.

C • News & context ("reading the internet")

Feature	Priority	Description
News ingestion	V1	Pulls headlines and articles relevant to the instrument from financial news sources.
Economic calendar	V1	Scheduled high-impact events (rate decisions, CPI, jobs data) flagged before they hit.
LLM news classification	V1	The model classifies each story (macro, regulatory, geopolitical, risk-off, hype) and scores likely impact and confidence.
Social sentiment	V2	Aggregate public sentiment as a low-weight, heavily-discounted input — it is noisy and easily manipulated.

D · Forecasting & signals

Feature	Priority	Description
Feature engineering	V1	Indicators (moving averages, RSI, MACD, ATR), lagged returns, distance from key levels, and news features.
Baseline rule model	V1	A simple, transparent benchmark every fancier model must beat — the honest yardstick.
ML forecast model	V1	Gradient boosting on tabular features, outputting a probability of an up/down move — never a point price.
Confidence + calibration	V2	A reliability dashboard checks that “60% confidence” signals actually win about 60% of the time.
Regime detection	V2	Flags when current conditions are unlike the training data and lowers confidence or abstains.
Model explainability	V2	Per-prediction feature attribution so the mentor can show which inputs drove the lean.

E · Risk-management engine

Feature	Priority	Description
Position-size calculator	MVP	Sizes trades from account risk %, stop distance, and pip value. The foundation, already prototyped.
ATR-based stop/target helper	V1	Suggests stop and target distances scaled to current volatility rather than guesswork.
R:R & expectancy display	MVP	Shows risk:reward and expected value on every hypothetical trade.
Account guardrails	V1	Max risk per trade, max simultaneous open risk, and a daily loss limit — each explained by the mentor.
Risk-of-ruin simulator	V2	Monte-Carlo simulation of trade sequences to show drawdown and probability-of-ruin under the chosen rules.
Multiple sizing methods	V2	Fixed-percent, fixed-ratio, and a cautious capped Kelly — with strong warnings on aggressive sizing.

F • Discipline & journal

Feature	Priority	Description
Trade journal	MVP	Logs every paper or live trade: entry, exit, size, the signal and confidence at the time, and the reason.
Pre-trade checklist	V1	A short checklist that must be completed before an entry — forcing a reason, a stop, and a target every time.
Mistake & emotion tagging	V1	Tag trades (moved stop, FOMO, revenge, tilt) to surface the behavioural patterns that cost money.
Personal analytics	V1	The user's own win rate, expectancy, and R-multiple distribution — the real measure of progress.
Trading-plan rules engine	V2	Define personal rules; the app checks trades against them and flags violations.

G • Backtesting & validation

Feature	Priority	Description
Event-driven backtester	V1	Replays signals and risk rules over history with no lookahead and full transaction-cost modelling.
Out-of-sample & walk-forward	V1	True held-out testing and rolling re-validation to expose overfitting.
Monte-Carlo sequence test	V2	Shuffles trade order thousands of times to reveal the realistic range of outcomes and drawdowns.
Strategy comparison	V1	Equity curves and metrics side by side, with the mentor flagging warning signs.

H · Alerts & workflow

Feature	Priority	Description
Price & event alerts	V1	Notifications on chosen price levels and approaching high-impact events.
Signal-change alerts	V2	Notify when the daily lean or the news read shifts materially.
Event freeze	V2	Warns before — or optionally blocks — new trades in the window around a high-impact release.

I · Dashboard & platform

Feature	Priority	Description
Unified dashboard	MVP	Price, lean, reasoning, news, upcoming events, open risk, and suggested size on one calm screen.
Annotated chart	V1	Interactive chart with indicators and the ability to mark levels and review past trades.
Prediction audit log	V1	Every forecast is logged against its real outcome, closing the calibration loop.
Single-user auth & secrets	MVP	One protected login; all API keys held server-side, never in the browser.

07 The mentor & curriculum

The “explain why” requirement is the heart of the product. It is delivered three ways: a built-in lesson tree, contextual explainers on every metric, and a reasoning expander on every signal. An `explain(metric, context)` service passes the actual current values to the language model so explanations are specific, not generic.

Built-in curriculum

1. **Market basics** — price, bid/ask/spread, lots, pips, leverage, margin.
2. **Risk first** — position sizing, stops and targets, R:R, the 1% rule, why risk beats prediction.
3. **Reading charts** — trend, support/resistance, candlesticks, taught skeptically as weak evidence rather than magic.
4. **Indicators & their failure modes** — moving averages, RSI, MACD, ATR, and where each misleads.
5. **Expectancy & the math of survival** — why a 40%-win system can still be highly profitable.
6. **Backtesting honestly** — lookahead bias, overfitting, transaction costs, out-of-sample discipline.
7. **Psychology & process** — journaling, tilt, and why the user’s own behaviour is the biggest risk.

EXAMPLE REASONING THE MENTOR PRODUCES

“Lean slightly long (58%, low confidence) because price is above the 200-day average and momentum is positive, but the spread is wide and a central-bank statement is due in three hours — which is why confidence is low. A move below yesterday’s low would flip this read.”

08 The forecasting pipeline

The language model handles **language and reasoning**; the quantitative model handles **numbers**. They are never swapped — an LLM has no reliable sense of price and must never produce a target.

1. **Ingest** the latest price across timeframes, plus news and the economic calendar.
2. **Engineer features** from price and from the LLM-scored news.
3. **Run the quantitative model** → probability of up/down over the horizon, with a confidence band.
4. **Regime check** → if conditions are abnormal, lower confidence or abstain.
5. **LLM synthesis** → turn the model output and news into a clear, uncertainty-honest narrative.
6. **Risk engine** → if the user chooses to act, size the trade from their risk rules.
7. **Display** the lean, the reasoning, the risks, and the “what would change my mind” factors.
8. **Log** the prediction against its eventual outcome to measure real-world accuracy over time.

09 System architecture

A clean separation of concerns with a strict one-way data flow. The outside world can feed data *in*; nothing in fetched content is ever treated as a command to the application. API keys live server-side only.

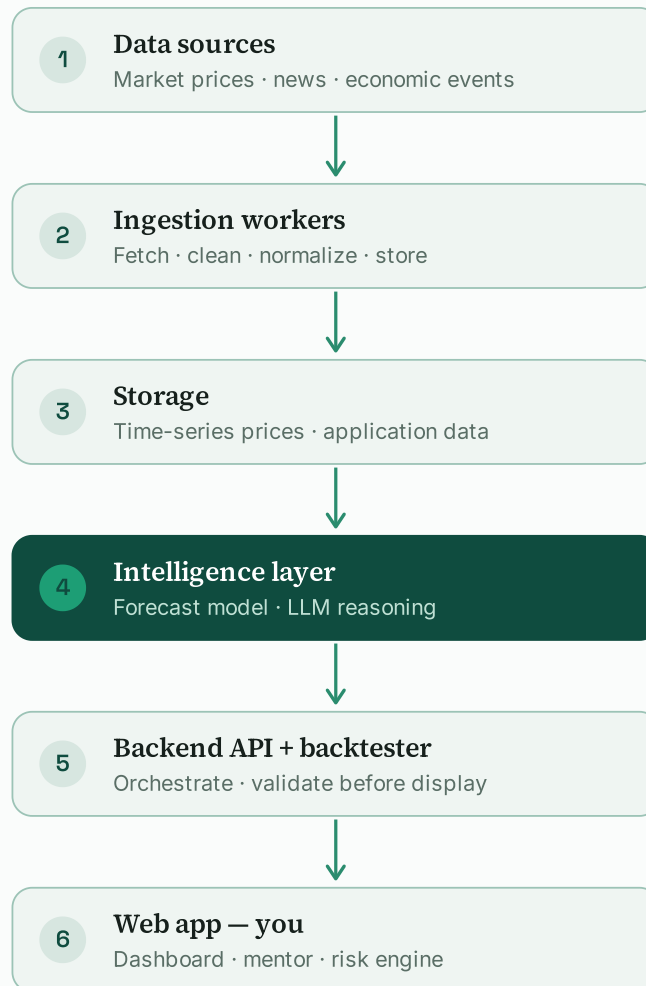


Figure 1 — One-way data flow from external sources through ingestion, storage, and the intelligence layer, validated by the backend before reaching the user.

- **Data ingestion workers** — scheduled jobs that fetch price, news, and the calendar, then clean, normalize, and store them.
- **Storage** — a time-series store for prices and indicators; an application database for settings, lessons, journal, predictions, and backtests.
- **Intelligence layer** — the quantitative forecast model and the LLM reasoning service, kept distinct by responsibility.
- **Backend API + backtester** — orchestrates everything and validates any signal before it can surface in the UI.
- **Web app** — the dashboard, mentor, charts, risk engine, and journal the user interacts with.

10 Data model

Key entities the application persists. Exact schemas evolve, but these relationships anchor the design.

Entity	Purpose	Key fields
Instrument	The tradable thing and its mechanics	symbol, pip size, lot size, quote currency
PriceBar	One OHLCV candle at a timeframe	instrument, timeframe, ts (UTC), o/h/l/c, volume
NewsItem	An ingested story with its LLM read	ts, source, headline, category, impact score, confidence
EconomicEvent	A scheduled calendar release	ts, name, importance, forecast, actual
Prediction	A logged forecast for calibration	ts, horizon, prob_up, confidence, reasoning, outcome
Trade	A paper or live position	entry, exit, size, direction, stop, target, R-multiple
JournalEntry	The user's reflection on a trade	trade, reason, mistake tags, emotion, notes
BacktestRun	A validated strategy result	strategy, period, costs, equity curve, metrics
Lesson / Progress	Curriculum and the user's standing	module, status, score, last reviewed
Alert	A user-defined trigger	type, condition, status, fired_at

11 Technology stack

Layer	Recommended choice	Why
Frontend	React + a charting library (Lightweight-Charts / Recharts) + Tailwind	Fast, component-driven UI; strong charting; single-page app fits a dashboard.
Backend	Python with FastAPI	Keeps backend and ML in one language; modern, async, well-documented.
ML & data	pandas, scikit-learn, XGBoost / LightGBM; PyTorch later	Proven tabular toolchain; deep learning only once basics work.
Indicators / backtest	pandas-ta; backtrader or vectorbt	Mature libraries for indicators and event-driven backtesting.
Reasoning / mentor	Anthropic API (Claude), server-side only	Strong reasoning for news classification and explanation; single-user makes cost trivial.
Databases	PostgreSQL with the TimescaleDB extension	One engine covers both application data and time-series prices.
Jobs / scheduling	Celery or cron-style scheduler	Drives ingestion and the daily model run.
Hosting	A single small cloud VM or container host	Ample for one user; keep secrets in environment variables.

12 External data sources

BEFORE YOU BUILD

Verify each provider's current pricing, free-tier limits, and terms of service before relying on it — these change, and ToS compliance matters when “reading from the internet.” Wrap every source behind an adapter so it can be swapped without touching the rest of the app.

Type	Candidate sources	Notes
Forex price	Twelve Data, Alpha Vantage, Polygon.io, broker APIs (e.g. OANDA)	History and live quotes; broker APIs give the most realistic pricing.
News	NewsAPI, Finnhub, Marketaux, financial RSS	Headlines and articles for the LLM to classify.
Economic calendar	Finnhub, Trading Economics, Forex Factory calendar	High-impact event scheduling and flags.
Sentiment (optional)	Public sentiment APIs where ToS permits	Low weight; noisy and manipulable.
Reasoning	Anthropic API	News reasoning and the mentor layer.

13 Backtesting & validation methodology

The backtester is the judge of everything. It exists to stop the user from trusting a strategy that only looks good because of a hidden flaw. Three failure modes must be designed out from the start.

THE THREE TRAPS

Lookahead bias — using future information to predict the past; the subtlest and most fatal bug.

Overfitting — a model that memorises historical noise and collapses live. **Ignoring costs** — spread, commission, and slippage quietly erasing a paper edge.

Defences built into the methodology: at every simulated step only past data is visible; strategies are evaluated on a held-out out-of-sample period and re-validated with walk-forward windows; transaction costs are modelled in full; and a Monte-Carlo pass shuffles trade order to reveal the realistic range of drawdowns rather than one lucky path. The product rule is absolute: **no signal reaches the dashboard until it has passed this process.**

14 Non-functional requirements

Quality	Requirement
Security	Single protected login; all API keys and secrets server-side; nothing sensitive in the browser or in URLs.
Privacy	Personal data stays local to the user's instance; the most privacy-preserving option is chosen by default.
Safety boundary	Fetches web content is data, never instructions. No external content can trigger an action; the human confirms every order.
Performance	Dashboard loads in under ~2s on cached data; the daily model run completes well within the trading-day schedule.
Reliability	Ingestion is idempotent and retries on failure; data-quality checks flag gaps rather than corrupting models silently.
Maintainability	Adapters isolate each data source; the instrument is configuration, not hard-coded.
Accessibility	Readable contrast, keyboard navigation, and a calm, uncluttered interface in both light and dark modes.

15 Risk register

Risk	Impact	Likelihood	Mitigation
Over-trusting the forecast	High	High	Honest confidence, calibration tracking, mentor framing, mandatory backtest proof.
Overfit backtest	High	High	Out-of-sample and walk-forward testing; Monte-Carlo; benchmark against the baseline.
Lookahead bias	High	Medium	Strict point-in-time data access; code review of the backtester; audit tests.
Costs erase the edge	Medium	High	Full spread/slippage/commission modelling in every backtest.
Data source change / outage	Medium	Medium	Adapter pattern, redundant sources, data-quality monitoring.
User psychology / tilt	High	Medium	Pre-trade checklist, guardrails, mistake tagging, journal review.
Prompt injection via news	Medium	Low	Treat all fetched content as data; the model never executes instructions found in it.

16 Roadmap & milestones

Phase	Focus	Key deliverables	Exit criteria
0 • Foundation	Risk engine	Position-size calculator; core risk concepts	Correct sizing and R:R on any inputs.
1 • Data + journal	Price & discipline	Ingestion, chart, trade journal with expectancy	The user can paper-trade and measure themselves.
2 • Mentor v1	Teaching	Lesson tree; contextual “explain this”	Every metric is explainable on demand.
3 • Backtester	Validation	Honest, cost-aware, out-of-sample engine	A strategy can be proven or disproven.
4 • Signals + news	Forecasting	Features, baseline + ML model, news + LLM read	A probabilistic read, each signal backtested first.
5 • Polish	Depth	Calibration, regime detection, Socratic mode, alerts	The complete, refined product.

BUILD ORDER MATTERS

Resist building Phase 4 first. Everyone does, and everyone regrets it — an unvalidated signal that looks brilliant on past data will take real money teaching the lesson the backtester would have given for free.

17 Success metrics

The product succeeds if the *user* improves — not if the forecast beats the market.

DISCIPLINE

100%

Trades with a pre-defined stop and target.

FORECAST CALIBRATION

Well-calibrated

Stated confidence matches realised hit rate.

PERSONAL EXPECTANCY

Positive & rising

The user's own edge, measured over many trades.

LEARNING

Curriculum complete

Lessons finished and retained via quizzes.

RISK CONTROL

No rule breaches

Guardrails respected; drawdown within plan.

ENGAGEMENT QUALITY

Journalled

Every trade reviewed, not just placed.

18 Cost estimate

Item	Estimate	Notes
Market & news data	Free tier → modest monthly	Free tiers often suffice for one instrument; paid only if outgrown.
LLM API	Low, pay-per-use	One user's news reads and explanations are inexpensive; a capable model is worth it.
Hosting	Low monthly	A single small VM or container host.
Your time	The real cost	Budget effort, not dollars — this is where the investment actually goes.

19 Compliance & responsible use

- A personal, educational tool — not licensed financial advice. The interface should say so plainly.
- Respect every data source's terms of service and rate limits.
- Paper-trade first. If a live broker is ever connected, require explicit human confirmation for every order and never let a model or a headline place a trade automatically.
- Keep all API keys server-side and out of version control.

20 Glossary

Term	Meaning
Pip	The smallest standard price increment of a currency pair (0.0001 for most majors).
Lot	A standardised trade size; one standard lot is 100,000 units (mini 10,000, micro 1,000).
Spread	The gap between the buy and sell price — a core trading cost.
Leverage / margin	Borrowed exposure and the collateral required to hold it; amplifies both gains and losses.
Stop loss / take profit	Pre-set exit prices for a losing and a winning trade.
Risk : reward (R:R)	Reward distance divided by risk distance; at 1:2 you can be wrong over half the time and still profit.
Expectancy	$(\text{win}\% \times \text{avg win}) - (\text{loss}\% \times \text{avg loss})$; the average profit per trade over many trades.
Drawdown	The peak-to-trough decline in account equity.
Risk of ruin	The probability of losing enough capital to be unable to continue.
Slippage	The difference between the expected and the actual fill price.
Lookahead bias	Accidentally using future data when testing on the past — inflates results fatally.
Overfitting	A model that fits historical noise and fails on new data.
Out-of-sample / walk-forward	Testing on data the model never saw, and rolling that test forward through time.
ATR / RSI / MACD	Common indicators for volatility, momentum strength, and momentum shifts.
Calibration	Whether a stated confidence matches the realised hit rate.
Regime	The prevailing market condition (e.g. trending vs choppy, calm vs volatile).

Build the teacher first. Let the predictions be humble. Let the risk engine be strict. That is how people who are still trading after fifteen years build things.